POINT REACTOR KINETICS EQUATIONS (PRKE) WITH FHR SIMULATOR

An Introduction to Reactor Dynamics

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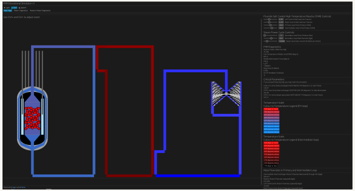
MISSION BRIEFING: GRID DEMAND INCREASE

Currently, our Fluoride-salt-cooled High-temperature Reactor (FHR) is operating stably, supplying baseload power to the grid.

The Situation: Grid demand has just spiked. The grid operator requests an immediate increase in power output.



Your Objective

Increase reactor thermal power from **30 MWth** to **60 MWth** and stabilize it there.

OPERATIONAL CONSTRAINTS & SAFETY LIMITS

Nuclear reactors are powerful systems that require precise control. We cannot just ”floor the accelerator.”

The Rules of Engagement:

- You have full control over the reactor’s control rods (reactivity).
- You must aim for exactly **60 MWth**.

SAFETY LIMIT (FAILURE CONDITION)

If reactor power exceeds **1000 MWth** at any point during the maneuver, you have violated the reactor’s safety limits and failed the mission.

YOUR INTERFACE: THE FHR CONTROL PANEL

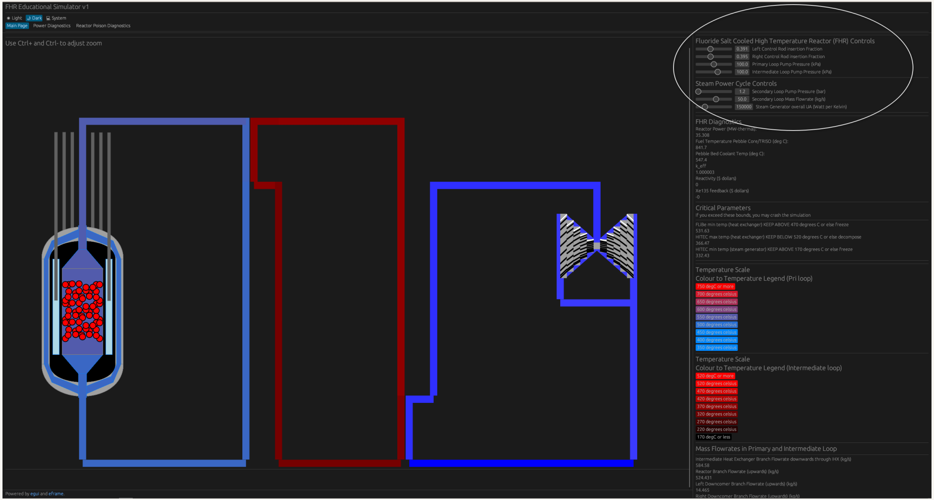


Figure: *Current State: Stable at 30 MWth*

Volunteers Needed!

Who thinks they have the steady hand to bring us to 60 MWth without exceeding 1000 MWth?

For the rest of the class (Observers): Watch closely. Don't just look at the power level.

- How does the reactor respond immediately after they move the controls?
- What happens to the power when they *stop* moving the controls?
- Does the reactor feel "heavy" (slow to move) or "twitchy" (fast to move)?

DEBRIEF: INTUITION VS. REALITY

We just saw a few attempts. It wasn't as easy as turning up a volume knob on a radio.

Discussion Questions:

1. Did anyone overshoot the 60 MWth target? Why did that happen?
2. Did you notice the power continuing to change even after you put the controls back to neutral?
3. If you added too much reactivity, did it feel like the reactor was "running away" from you?

Why does the reactor have "momentum"? Why doesn't it stop instantly?
To answer this, we need to look at the physics of neutron populations.

THE HIDDEN MECHANISM

You experienced the reactor's behavior:

- It doesn't respond instantly to control rod changes.
- It has "momentum" – power keeps changing after you stop acting.
- If you add too much reactivity quickly, it feels uncontrollable.

To understand why, we need to look at the life cycle of neutrons.

We use the **Point Reactor Kinetics Equations (PRKE)** to model this.

STEP 1: THE NEUTRON BALANCE

At its simplest, reactor power changes based on the population of neutrons (n) Bell and Glasstone 1970.

$$\text{Rate of Change} = \text{Production} - \text{Loss}$$

$$\frac{dn}{dt} = \text{Production} - \text{Loss}$$

The production and loss terms depend on many nuclear processes:

- Fission reactions (production)
- Absorption in fuel, coolant, structure (loss)
- Leakage out of the core (loss)
- Fast vs thermal neutron behavior

How do we capture all these effects in one framework?

THE SIX FACTOR FORMULA

Nuclear engineers use the **six factor formula** to account for all the ways neutrons are gained and lost in a reactor Lamarsh, Baratta, et al. 2001:

$$k_{eff} = \underbrace{P_{TNL} \cdot P_{FNL}}_{\text{Non-Leakage}} \cdot \underbrace{\eta}_{\text{Reproduction}} \cdot \underbrace{\epsilon}_{\text{Fast Fission}} \cdot \underbrace{f}_{\text{Thermal Utilization}} \cdot \underbrace{p}_{\text{Resonance Escape}}$$

Where:

- P_{TNL}, P_{FNL} = Probability of not leaking (thermal and fast)
- η = Neutrons produced per neutron absorbed in fuel
- ϵ = Fast fission factor
- f = Thermal utilization factor
- p = Resonance escape probability

WHAT DOES k_{eff} TELL US?

The effective multiplication factor k_{eff} represents the ratio of neutrons in one generation to the next:

$$k_{eff} = \frac{\text{Neutrons in Generation } (n+1)}{\text{Neutrons in Generation } n} = \frac{\text{Production}}{\text{Loss}}$$

This leads to three reactor states:

- $k_{eff} < 1$: **Subcritical** – neutron population decreasing
- $k_{eff} = 1$: **Critical** – neutron population constant
- $k_{eff} > 1$: **Supercritical** – neutron population increasing

Now we can rewrite our balance equation:

$$\frac{dn}{dt} = \text{Loss} \cdot (k_{eff} - 1)$$

STEP 2: WHAT IF ALL NEUTRONS WERE FAST?

Imagine every neutron is born instantly after fission (”prompt neutrons”).
The ”Loss” term depends on how quickly neutrons are removed. We define the **neutron lifetime** (l) as the average time a neutron survives before being lost (absorbed or leaked) Lamarsh, Baratta, et al. 2001.

$$\text{Loss Rate} = \frac{n}{l}$$

Substituting this into our balance equation gives the **Prompt Kinetics Equation**:

$$\frac{dn}{dt} = \frac{n}{l}(k_{eff} - 1)$$

This equation assumes *instantaneous* response.

HOW FAST ARE PROMPT NEUTRONS BORN?

Prompt neutrons are emitted virtually instantaneously after a fission event occurs.

Timescale for prompt neutron emission Mihalczo 2004:

$$t_{\text{prompt emission}} \approx 10^{-14} \text{ seconds}$$

This is **10 femtoseconds** – essentially instantaneous on any human or engineering timescale.

Once born, these neutrons then live for a time period l (the neutron lifetime) before being absorbed or leaking out. This lifetime l determines how quickly the reactor responds.

THE PROBLEM WITH PROMPT NEUTRONS: SPEED

How fast is this response?
The neutron lifetime l depends on the reactor type Lamarsh, Baratta, et al. 2001:

Reactor Type	Neutron Energy	Neutron Lifetime (l)
Thermal (LWR, FHR)	$\sim$0.025 eV	10^{-4} to 10^{-5} s
Fast (SFR, metal core)	$\sim$1 MeV	10^{-7} to 10^{-8} s

Fast reactors respond 1000× faster than thermal reactors!

If $k_{eff} = 1.001$ in a fast reactor, power doubles in microseconds.

THE PROBLEM WITH PROMPT NEUTRONS: SPEED

Reality Check

Human reaction time $\approx$ 0.25 seconds = 250,000 microseconds.
Prompt-only reactors would be **impossible** for humans to control.

STEP 3: NATURE'S SAFETY BRAKE - DELAYED NEUTRONS

Thankfully, not all neutrons are born instantly Bell and Glasstone 1970.
A small fraction, denoted by β (beta), are emitted by fission fragments (called **precursors**, C) seconds or even minutes later.

- **Prompt Neutrons ($1-\beta$):** Appear instantly ($\sim 10^{-14}$ s).
- **Delayed Neutrons (β):** Appear later, governed by the decay of precursors (seconds to minutes).

This delay is what creates the "momentum" or lag you felt in the simulator. The precursors act like a battery, storing potential neutrons and releasing them slowly.

THE POINT REACTOR KINETICS EQUATIONS (1-GROUP)

To model reality, we need two coupled equations: one for neutrons (n) and one for precursors (C) Bell and Glasstone 1970.
We introduce two key parameters:

- **Reactivity (ρ):** $\rho \equiv \frac{k_{eff}-1}{k_{eff}}$ (dimensionless measure of departure from criticality)
- **Generation Time (Λ):** $\Lambda = \frac{l}{k_{eff}}$ (average time between successive generations of neutrons)

THE POINT REACTOR KINETICS EQUATIONS (1-GROUP)

The 1-Group PRKE (simplified):

$$\underbrace{\frac{dn}{dt}}_{\text{Change in Power}} = \underbrace{\frac{\rho - \beta}{\Lambda} n}_{\text{Prompt Term}} + \underbrace{\lambda C}_{\text{Delayed Source}}$$

$$\underbrace{\frac{dC}{dt}}_{\text{Change in Precursors}} = \underbrace{\frac{\beta}{\Lambda} n}_{\text{New Precursors}} - \underbrace{\lambda C}_{\text{Precursor Decay}}$$

REALITY: SIX DELAYED NEUTRON PRECURSOR GROUPS

In practice, delayed neutrons come from many different fission products with different decay rates. We typically group them into **6 precursor groups** Bell and Glasstone 1970.

The 6-Group PRKE:

$$\frac{dn}{dt} = \frac{\rho - \beta}{\Lambda} n + \sum_{i=1}^{6} \lambda_i C_i$$

$$\frac{dC_i}{dt} = \frac{\beta_i}{\Lambda} n - \lambda_i C_i \quad \text{for } i = 1, 2, ..., 6$$

Where $\beta = \sum_{i=1}^{6} \beta_i$ is the total delayed neutron fraction.

Each group has different:

REALITY: SIX DELAYED NEUTRON PRECURSOR GROUPS

- Delayed neutron fraction (β_i)
- Decay constant (λ_i) or half-life ($t_{1/2,i} = \ln(2)/\lambda_i$)

DELAYED NEUTRON DATA BY FUEL TYPE

Decay constants (same for all fuels):

Group	Half-life (s)	Decay constant λ (s^{-1})
1	56.0	0.0124
2	23.0	0.0301
3	6.2	0.1118
4	2.3	0.3013
5	0.61	1.136
6	0.23	3.013

Delayed neutron fractions β_i (fuel-dependent):

DELAYED NEUTRON DATA BY FUEL TYPE

Group	U-233	U-235	Pu-239
1	0.00023	0.00021	0.00007
2	0.00078	0.00142	0.00063
3	0.00064	0.00128	0.00044
4	0.00074	0.00257	0.00069
5	0.00014	0.00075	0.00018
6	0.00008	0.00027	0.00009
Total β	**0.00261**	**0.0065**	**0.0021**

WHY SIX GROUPS MATTER

Why not just use one average group?

Different Time Scales

- **Fast groups (5, 6):** Half-lives $<$ 1 second — respond quickly to power changes
- **Intermediate groups (3, 4):** Half-lives 2-6 seconds — provide medium-term stability
- **Slow groups (1, 2):** Half-lives 23-56 seconds — provide long-term ”memory”

WHY SIX GROUPS MATTER

Practical Impact

- Fast groups dominate initial response to reactivity changes
- Slow groups determine long-term equilibrium behavior
- One-group approximation is acceptable for quick estimates
- Six-group model required for accurate transient simulations

Your FHR simulator uses the full 6-group model for accurate dynamics!

THE EDGE OF CONTROL: PROMPT CRITICALITY

Look closely at the prompt term in the neutron equation:

$$\frac{\rho - \beta}{\Lambda} n$$

Normal operation involves adding small amounts of reactivity ($\rho < \beta$). The prompt term is negative, and the reactor relies on the delayed source ($\sum \lambda_i C_i$) to increase power slowly.

What happens if you add so much reactivity that $\rho > \beta$?

Prompt Critical

The prompt term becomes **positive**. The reactor power begins to rise exponentially based *only* on prompt neutrons, on a timescale of microseconds.
Control is lost instantly.

UNITS OF REACTIVITY: THE DOLLAR

Because the fraction β is the boundary between controllable and uncontrollable, we use it as a unit.

$$\text{Reactivity in Dollars (\$)} = \frac{\rho}{\beta}$$

- **\$0.10 (10 cents):** A typical, safe control movement.
- **\$0.50 (50 cents):** A large, but manageable insertion.
- **\$1.00 (Prompt Critical):** The exact boundary where $\rho = \beta$.
- **> \$1.00 (Super-Prompt Critical):** The danger zone.

For typical reactors: $\beta \approx 0.0065$ (U-235) to 0.0021 (Pu-239).
So \$1.00 $\approx$ 650 pcm (percent-mille) to 210 pcm of reactivity.

Now that you understand reactivity in dollars, let's try the power maneuver again.

Updated Mission: 30 MWth → 60 MWth

This time, the FHR simulator displays **reactivity in dollars ($)** on the interface.

New constraint: Never exceed **$0.50** during the maneuver.

Observers: Watch for:

- How much reactivity does the operator insert?
- How does the reactor respond at different reactivity levels?
- What happens when reactivity returns to zero?

Who wants to try with this new information?

WAIT... WHY DOESN'T POWER GO TO INFINITY?

Looking at our PRKE equation:

$$\frac{dn}{dt} = \frac{\rho - \beta}{\Lambda} n + \sum_{i=1}^{6} \lambda_i C_i$$

Question: If we keep reactivity positive ($\rho > 0$), shouldn't power keep increasing forever?

In your simulator attempts, you probably noticed:

- Power eventually stabilizes even with positive reactivity inserted
- The reactor seems to "push back" as power increases
- Temperature rises as power increases

What's missing from our equations?

REACTIVITY FEEDBACK: THE REACTOR RESPONDS TO ITSELF

In reality, reactivity ρ is **not constant**—it changes with reactor conditions Bell and Glasstone 1970; Lamarsh, Baratta, et al. 2001.

Key insight: As power increases, temperature increases. Temperature changes affect neutron cross sections and material densities, which change k_{eff}, which changes ρ.

We can write:

$$\rho(t) = \rho_{\text{external}} + \rho_{\text{feedback}}(T_{\text{fuel}}, T_{\text{coolant}}, \ldots)$$

Where:

- ρ_{external} = reactivity from control rods (what you control)
- ρ_{feedback} = reactivity changes from temperature, density, etc.

The most important feedback mechanisms are temperature-related Lamarsh, Baratta, et al. 2001:

Fuel Temperature Coefficient (α_F):

$$\rho_{\text{fuel}} = \alpha_F (T_{\text{fuel}} - T_{\text{fuel,0}})$$

Coolant Temperature Coefficient (α_C):

$$\rho_{\text{coolant}} = \alpha_C (T_{\text{coolant}} - T_{\text{coolant,0}})$$

These coefficients can be:

- **Negative** ($\alpha < 0$): Temperature increase $\rightarrow$ reactivity decreases (stabilizing)
- **Positive** ($\alpha > 0$): Temperature increase $\rightarrow$ reactivity increases (destabilizing)

Most reactors are designed with **negative temperature coefficients** Bell and Glasstone 1970.

How it works:

1. You add positive reactivity $\rightarrow$ power increases
2. Power increase $\rightarrow$ fuel temperature increases
3. Fuel temperature increase $\rightarrow$ negative feedback reduces reactivity
4. Reactivity approaches zero $\rightarrow$ power stabilizes

This is why power doesn't go to infinity!

The reactor has a built-in ”thermostat” that automatically opposes changes. This is called **inherent safety** or **passive safety**.

This is what you observed in the simulator—power eventually leveled off even when you kept control rods in place.

Zero-Power PRKE (what we derived earlier):

$$\frac{dn}{dt} = \frac{\rho - \beta}{\Lambda} n + \sum_{i=1}^{6} \lambda_i C_i$$

Assumes ρ depends only on control rod position. Valid for low power where temperature changes are negligible.

Full-Power PRKE (with feedback) Bell and Glasstone 1970:

$$\frac{dn}{dt} = \frac{\rho(T) - \beta}{\Lambda} n + \sum_{i=1}^{6} \lambda_i C_i$$

$$\rho(T) = \rho_{\text{rods}} + \alpha_F \Delta T_F + \alpha_C \Delta T_C$$

ZERO-POWER VS FULL-POWER KINETICS

Must solve coupled neutronics + thermal-hydraulics equations.

The FHR simulator you've been using includes simplified feedback models, which is why you see realistic stabilization behavior.

ADVANCED CONTROL: LOAD FOLLOWING WITHOUT CONTROL RODS

Traditional approach: Adjust control rods to change power output.

Modern approach: Use temperature feedback for automatic load following Zhang and Jiang 2024.

How It Works

When grid demand changes, adjust coolant flow rate instead of control rods:

- **Increase demand:** Increase coolant flow $\rightarrow$ lower coolant temp $\rightarrow$ less negative feedback $\rightarrow$ power rises
- **Decrease demand:** Decrease coolant flow $\rightarrow$ higher coolant temp $\rightarrow$ more negative feedback $\rightarrow$ power drops

Why is this viable for advanced SMRs?

- **TRISO fuel:** Much higher thermal margins (~1600°C) compared to traditional UO_2 with zircalloy cladding (~800°C, strong degradation via oxidation Duriez et al. 2019)
- Temperature excursions during load following stay well within safe limits
- Strong negative temperature coefficients enable stable, passive control

TURBINE THROTTLE CONTROL

PWR Traditional Approach (Submarines and other types): Turbine bypass valve Zhang and Jiang 2024

How It Works

When power demand changes, adjust coolant flow rate instead of control rods:

- **Increase demand:** Decrease flow to turbine, increase flow to bypass valve
- **Decrease demand:** Increase flow to turbine, decrease flow to bypass valve

In essence, exhaust the steam from the steam generator directly to cool rather than generate useful work output. This is rather wasteful, but it works for temporary load following too.

CRITICAL: FHR SALT TEMPERATURE LIMITS

Unlike water-cooled reactors, FHR salts have strict temperature operating windows.

Primary Loop: FLiBe (Lithium Fluoride - Beryllium Fluoride)

Minimum temperature: $> 470°C$

- Below 470°C: FLiBe **freezes** (melting point $\sim$ 459°C)
- Frozen salt blocks circulation passages
- Loss of cooling capability

CRITICAL: FHR SALT TEMPERATURE LIMITS

Secondary Loop: HITEC Salt (Potassium/Sodium/Sodium Nitrate)

Operating range: 170°C to 520°C

- Below 170°C: HITEC **freezes** (melting point $\sim$ 142°C)
- Above 520°C: HITEC begins to **thermally decompose**
- Decomposition produces gases, damages salt chemistry

Important: The simulator will crash if these limits are violated—it cannot model phase change or salt decomposition!

SIMULATOR EXERCISE: POWER REDUCTION VIA PUMP SHUTDOWN

Mission: Reduce power from 60 MWth to 30 MWth by shutting down one or more coolant pumps.

Rules

- **DO NOT** move control rods during the maneuver
- Keep reactivity in dollars within ± 0.30 at all times
- Control power *only* by shutting down pumps (reducing coolant flow)
- Monitor reactor period and ensure it remains > 20 seconds
- **CRITICAL:** Keep FLiBe > 470°C and HITEC within 170-520°C

Strategy hints:

SIMULATOR EXERCISE: POWER REDUCTION VIA PUMP SHUTDOWN

1. Start at stable 60 MWth operation
2. Check current salt temperatures—ensure adequate margins
3. Shut down one pump and observe the response
4. Watch coolant temperature rise (but not too much!)
5. Monitor how reactivity responds to temperature changes
6. If temperatures approach limits, stabilize or reverse action

This demonstrates passive power reduction—but with real engineering constraints!

WHAT YOU SHOULD OBSERVE

Phase 1: Flow Reduction

- Pump shuts down
- Coolant flow rate decreases
- Salt outlet temperature rises
- Core inlet temperature rises

Phase 2: Feedback Response

- Higher salt temperature $\rightarrow$ Higher Fuel Temperature
- Reactivity becomes more negative
- Power begins to decrease
- Fuel temperature drops

Phase 3: New Equilibrium

- Power stabilizes at lower level ($\sim$30 MWth)
- Temperatures reach new steady state
- Reactivity returns close to zero
- **Verify:** All salt temps within limits!

Key Insight

With proper negative feedback, the reactor is **self-regulating**. But you must respect salt temperature constraints Zhang and Jiang 2024.

WHY SALT TEMPERATURE LIMITS MATTER

Real-world operational challenges for molten salt reactors:

Freezing Scenario (Temperature Too Low)

- Salt solidifies in pipes, heat exchangers, or reactor vessel
- Blockage prevents coolant circulation
- Cannot restart reactor until salt is remelted
- Requires external heating systems (trace heaters)
- Can take hours or days to recover

WHY SALT TEMPERATURE LIMITS MATTER

Overheating Scenario (Temperature Too High)

- HITEC decomposition produces gases (NO_x, O_2)
- Pressurization of secondary loop
- Degraded heat transfer properties
- Corrosion acceleration
- Requires salt replacement (expensive, time-consuming)

Lesson: Flow-based control in FHRs must account for salt chemistry constraints, not just neutronics!

WHY NOT USE THIS IN LARGE COMMERCIAL PWRS?

Question: If pump control works so well, why don't large PWRs use it for load following?

WHY NOT USE THIS IN LARGE COMMERCIAL PWRS?

Safety and Design Constraints in Large PWRs

- **Pump trip is an accident scenario:** Loss of coolant flow is considered a design-basis accident in large PWRs. Deliberately reducing flow contradicts safety philosophy.
- **DNB (Departure from Nucleate Boiling) concerns:** Reduced flow at high power can lead to local boiling on fuel surfaces, risking fuel damage.
- **Thermal stratification:** Reduced flow can cause temperature stratification in the reactor vessel and steam generators, creating thermal stresses.
- **Pump wear:** Frequent pump speed changes or shutdowns accelerate mechanical wear on expensive, safety-critical equipment.
- **Large thermal inertia:** Large PWRs have massive coolant inventory and thermal mass, making flow-based control sluggish.

WHEN IS FLOW-BASED CONTROL ACCEPTABLE?

Flow-based load following is viable in certain reactor designs Zhang and Jiang 2024:

Small Modular Reactors (SMRs)

- Smaller thermal mass $\rightarrow$ faster response
- Higher surface-to-volume ratio $\rightarrow$ better heat removal margins
- Designed from the start for flexible operation
- Strong negative temperature coefficients
- Natural circulation capability reduces pump dependence

WHEN IS FLOW-BASED CONTROL ACCEPTABLE?

Fluoride-Salt-Cooled High-Temperature Reactors (FHRs)

- Very high temperature margins (no DNB risk)
- Liquid salt coolant doesn't boil under normal conditions
- Strong negative temperature feedback
- Designed for passive safety and load following
- **But:** Must respect salt temperature windows (freezing/decomposition)

Key takeaway: Reactor design determines which control strategies are safe and practical.

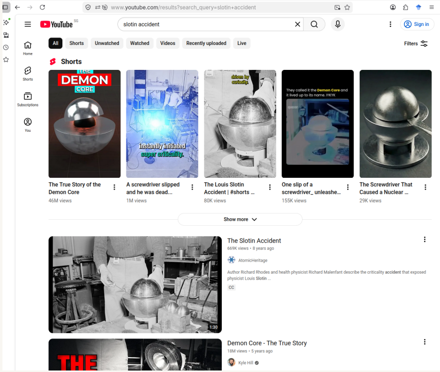
HISTORICAL WARNING: THE SLOTIN INCIDENT (1946)

Los Alamos National Laboratory, May 21, 1946
Stratton and Smith 1989.

Physicist Louis Slotin was demonstrating a criticality experiment to his replacement. The experiment involved bringing two hemispheres of plutonium (the "Demon Core") close together.

The only thing keeping them apart? A screwdriver blade.

The procedure was unauthorized and extremely dangerous. Colleagues called it "tickling the dragon's tail."



YouTube Videos of the "Demon Core" experiment setup.

At 3:20 PM, Slotin's screwdriver slipped Stratton and Smith 1989; Oettingen 2018.

The hemispheres came together. The core went **super-prompt critical** instantly—estimated at roughly **$1.10** Oettingen 2018.

What happened in that instant:

- Bright flash of blue light (Not Cherenkov radiation, but ionization)
- Wave of heat across Slotin's hand
- Sour taste in his mouth (ionization of air)
- Duration: less than one second

THE SLOTIN INCIDENT: THE ACCIDENT

Slotin immediately knocked the top hemisphere away with his bare hand, stopping the reaction. His quick action likely saved the lives of the seven other people in the room.

But the damage was done.

Radiation dose received: Estimated 2,100 rad (21 Gy) in less than one second Stratton and Smith 1989.

Timeline of Louis Slotin's condition:

- **Day 1:** Nausea and vomiting began within hours
- **Day 2-3:** Severe radiation burns appeared on his hands
- **Day 4-6:** Mental confusion, loss of coordination
- **Day 7-8:** Internal organs failing, extreme pain
- **Day 9:** Louis Slotin died on May 30, 1946

The Critical Lesson

Human reaction time ($\sim$0.25 s) is meaningless when prompt criticality unfolds in microseconds. Even Slotin's heroic reflexes couldn't save him—he had already received a lethal dose before his brain could process what happened.

After incidents like Slotin's, the nuclear community needed to understand prompt criticality in *reactor* systems, not just bare cores.

The Question: What happens if a water-cooled reactor goes prompt critical?

The Boiling Water Reactor Experiments (BORAX) at Idaho National Laboratory were designed to answer this—by deliberately destroying reactors.

The BORAX-I Final Test (July 22, 1954)

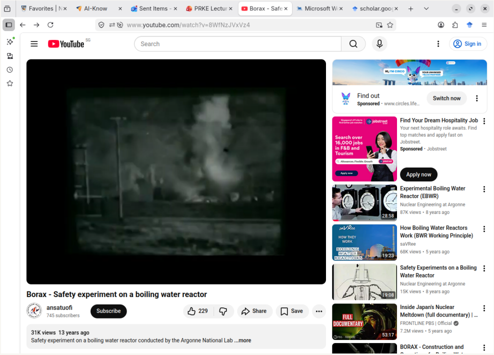
BORAX-I was built partially underground and operated remotely from a considerable distance. Scientists programmed the reactor to rapidly eject a high-worth control rod, inserting reactivity well above \$1.00 Putman 2012.
They evacuated everyone to a safe distance and filmed what happened.

Expected outcome Putman 2012:

- Severe core damage
- Fuel melting
- Release of ∼80 megajoules
- Relatively large steam explosion

Actual outcome:

- Released ∼**135 megajoules** (69% more!)
- Much more destructive than predicted
- Classified as an *accident* due to unexpected severity



BORAX-I final destructive test - YouTube Video

BORAX-I: SEQUENCE OF DESTRUCTION

What happened Putman 2012:

1. **Control rod ejected** – Rapid reactivity insertion (>$1.00)
2. **Nuclear excursion** – Power rose exponentially in milliseconds
3. **Extensive fuel melting** – Direct result of prompt criticality
4. **Steam explosion** – Initiated by damage from the criticality
5. **Massive destruction:**
    - 2,200 lb control rod mechanism thrown 30 feet into the air
    - Fuel pieces ejected up to 200 feet away
    - Fortunately, damage was limited to the reactor itself

Watch the BORAX destruction test (YouTube)
Read the full INL report (p. 117)

BORAX: WHY DIDN'T SAFETY FEATURES WORK?

Boiling Water Reactors (BWRs) have multiple passive safety features:

- Natural circulation cooling
- Negative void coefficient (steam bubbles reduce reactivity)
- Negative fuel temperature coefficient

So why did the reactor explode?

BORAX: WHY DIDN'T SAFETY FEATURES WORK?

The Speed Problem

Prompt criticality happens in **milliseconds**. The fuel temperature feedback *did* activate and stop the nuclear reaction—but not before the fuel temperature rose so high that it **melted the fuel**.

The molten fuel then violently interacted with water, causing a steam explosion that was **69% more energetic than predicted**.

Physics worked—but too slowly to prevent catastrophic damage.

CAN MODERN REACTORS SURVIVE PROMPT CRITICALITY?

It depends on the reactor design and fuel type.

Light Water Reactors (LWRs) / BWRs

Negative fuel temperature coefficient stops the reaction quickly.
However, the fuel temperature can rise so high (>2800°C) that the fuel **melts** before the feedback stops the excursion.
Result: Fuel damage, potential steam explosion (as in BORAX-I Putman 2012).

Key issue: Low thermal margin in UO_2 fuel—melting point is relatively close to operating temperature.

HIGH-TEMPERATURE REACTORS: BETTER MARGINS

HTGRs (High-Temperature Gas Reactors) and FHRs

These reactors use **TRISO fuel particles**—ceramic-coated fuel designed to withstand extreme temperatures (~1600°C).

Key advantages:

- **Higher thermal margins:** TRISO fuel can tolerate much higher temperatures before failure
- **Strong negative temperature coefficient:** Fuel temperature rise quickly reduces reactivity
- **Lower excess reactivity:** Less potential for large reactivity insertions

Result: Fuel temperature feedback *may* stop a prompt criticality event without damaging the fuel—**provided the excess reactivity insertion isn't too large.**

HIGH-TEMPERATURE REACTORS: BETTER MARGINS

This is why HTGRs and FHRs are considered to have superior passive safety characteristics.

THE BOTTOM LINE: NEVER GO PROMPT CRITICAL

Regardless of reactor type, prompt criticality is unacceptable.

- **LWRs/BWRs:** Prompt criticality leads to fuel melting and potential steam explosions—BORAX-I proved this catastrophically Putman 2012
- **HTGRs/FHRs:** May survive small prompt criticality events, but only if excess reactivity is limited
- **Fast reactors:** Even more sensitive due to shorter neutron lifetimes

The Engineering Rule

All reactors are designed with **reactivity limits** to ensure they can *never* reach $1.00 under any credible scenario.
Multiple independent systems (control rods, shutdown systems, reactivity coefficients) provide defense-in-depth.

This is why you must understand and respect the dollar boundary.

CASE STUDY 3: THE CHERNOBYL DISASTER (1986)

We've seen two examples of prompt criticality:

- **Slotin:** Bare critical assembly - instant death
- **BORAX:** BWR with negative feedback - still destroyed

What about a reactor with positive feedback?

On April 26, 1986, Reactor 4 at the Chernobyl Nuclear Power Plant in Ukraine experienced a catastrophic accident during a turbogenerator safety test Malko 2002.

The Result

The reactor went prompt critical with **positive feedback** amplifying the excursion. The explosion completely destroyed the reactor and released massive amounts of radioactive material across Europe, contaminating vast areas for decades.

THE RBMK REACTOR: A FLAWED DESIGN

The Chernobyl reactor was an RBMK-1000 (Reaktor Bolshoy Moshchnosti Kanalnyy) - a graphite-moderated, water-cooled channel reactor Malko 2002.

Design characteristics:

- Graphite moderator (slows neutrons)
- Water coolant (removes heat)
- Large physical size (11.8 m diameter, 7 m high core)
- Nominal thermal power: 3,200 MWth
- Positive steam-void coefficient at low power

THE RBMK REACTOR: A FLAWED DESIGN

The Fatal Flaw: Positive Steam-Void Coefficient

In most reactors (LWRs, BWRs), water acts as both coolant *and* moderator. Steam bubbles (voids) reduce moderation $\rightarrow$ reactivity decreases (negative feedback).

In RBMK, graphite is the moderator. Steam voids remove neutron-absorbing water $\rightarrow$ reactivity **increases** (positive feedback)! The coefficient could reach up to 5β at certain conditions.

UNDERSTANDING POSITIVE STEAM-VOID COEFFICIENT

Normal reactor (negative void coefficient):

1. Power increases $\rightarrow$ water heats up $\rightarrow$ steam forms
2. Steam has lower density $\rightarrow$ less moderation
3. Less moderation $\rightarrow$ fewer thermal neutrons $\rightarrow$ power decreases (stabilizing)

RBMK reactor (positive steam-void coefficient):

1. Power increases $\rightarrow$ water heats up $\rightarrow$ steam forms
2. Steam has lower density $\rightarrow$ less neutron absorption in water
3. Less absorption $\rightarrow$ more neutrons available $\rightarrow$ power **increases further** (destabilizing)

UNDERSTANDING POSITIVE STEAM-VOID COEFFICIENT

This creates a **positive feedback loop**: more power $\rightarrow$ more steam $\rightarrow$ more power $\rightarrow$ more steam...

The RBMK's positive steam-void coefficient was particularly dangerous at low power operation Malko 2002.

The Setup Malko 2002:

- Turbogenerator test during planned shutdown
- Power reduced: 3,200 MW $\rightarrow$ 1,500 MW $\rightarrow$ target 720 MWth
- ECCS switched off at 14:00 on April 25

Critical Mistake (00:28, April 26):

- Power dropped uncontrollably to 30 MWth
- Strong xenon poisoning occurred
- Deputy chief engineer Dyatlov **ordered** operators to withdraw nearly all control rods to raise power
- Operative reactivity surplus: only 6-8 rods

At 14 hr, the emergency core cooling system (ECCS) was switched off according to the experiment program. At this time, the Kiev dispatcher of the electrical grid ordered to continue the operation of the Unit 4 because of a shortage of power [4]. From this time the reactor was operated at the power 1,500 MW thermal with the switched-off ECCS system. The order of the Kiev distributor caused an important disturbance for the test because the later continuation of the experiment had to be done by another shift of the Unit 4 that was not planned for this important work.

At 23 hr 10 min, the operator of the reactor was allowed to decrease the power of the unit. At 00 hr 10 min on 26 April, it reached the level 720 MW thermal that was the lower limit of the power according to the experiment program [4]. However, the operator could not stabilize the power of the reactor at this level. It continues to fall and at 00 hr 28 min, it fell down to 30 MW thermal. The operator of the reactor, the senior engineer, Leonid Toptunov and the shift foreman, Alexander Akimov decided to insert absorbing rods in the core in order to shutdown the reactor. They were forced by the deputy chief engineer for operation of Unit 3 and Unit 4, Aleksander Dyatlov to withdraw the absorbers out the core in order to increase the power of the reactor [11]. The latter wished to carry out the planned test at any price. The necessity of the withdrawal of practically all absorbers out the core was dictated by a very strong xenon poisoning as a result of very quick decrease of the reactor power.

After the withdrawal of a number of absorbe... of the reactor began to increase. At 01 hr

Control rod withdrawal beyond safety limits Malko 2002

THE FATAL SEQUENCE (01:23, APRIL 26, 1986)

With nearly all control rods withdrawn, the reactor was in an extremely dangerous state Malko 2002:

01:23:04 - Test initiated:

1. Turbine valves closed $\rightarrow$ coolant flow decreased slightly
2. Small power increase began
3. Power increase $\rightarrow$ steam formation $\rightarrow$ **positive void feedback activated**
4. Reactivity increased rapidly $\rightarrow$ power surged uncontrollably

01:23:40 - SCRAM attempted (36 seconds later):

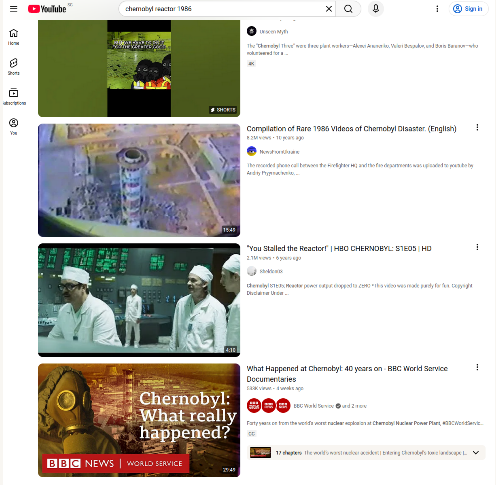
- Operators pressed AZ-5 (emergency shutdown) button
- All control rods began inserting into core
- **Design flaw:** Graphite displacers at rod tips initially *added positive reactivity* ("positive reactivity surge")

- Combined with positive void coefficient $\rightarrow$ catastrophic power excursion

01:23:47 - Destruction (6-7 seconds after SCRAM): Reactor destroyed by explosions before rods could fully insert.

The Explosions Malko 2002:

- At least two explosions occurred
- First: likely steam explosion
- Second: possibly nuclear explosion (estimated 200 tons TNT equivalent)
- Core completely destroyed
- 2,000 ton upper structure thrown upward
- Graphite fire released radioactive material for 10 days



Chernobyl Reactor 4 after the explosion.

[Read detailed analysis (Malko, 2002)]

The RBMK had negative fuel temperature feedback—why didn't it stop the accident?

Positive Feedback Overwhelmed Negative Feedback

The positive steam-void coefficient *amplified* the excursion faster than the negative fuel temperature coefficient could compensate.

The sequence:

1. Power increase $\rightarrow$ steam formation $\rightarrow$ **positive void feedback** (very fast)
2. Positive feedback $\rightarrow$ more power $\rightarrow$ more steam $\rightarrow$ runaway loop
3. Fuel temperature rises $\rightarrow$ negative fuel feedback activates (too slow)
4. By the time negative feedback became significant, the reactor was already physically destroyed

WHY FEEDBACK DIDN'T SAVE CHERNOBYL

Lesson: Positive feedback mechanisms can overpower negative ones if they act faster. This is why positive void coefficients are fundamentally unacceptable in reactor design.

LESSONS FROM CHERNOBYL

Design Lessons Malko 2002:

- **Never design reactors with positive void/steam coefficients**
- All modern reactors require negative reactivity coefficients
- Control rods must always add negative reactivity when inserted (no ”positive reactivity surge”)
- Adequate shutdown margin must be maintained at all times
- Proper documentation and understanding of reactor limitations essential

Operational Lessons:

- Safety systems must not be disabled during tests
- Operators must understand reactor physics (especially feedback mechanisms)
- Low-power operation in unstable configurations is extremely dangerous

LESSONS FROM CHERNOBYL

- Nuclear safety culture must prioritize safety over completing tests

The Common Thread

Slotin, BORAX, and Chernobyl all demonstrate: **prompt criticality is catastrophic**. Even ”safety features” like feedback can make things worse if they’re positive instead of negative.

THREE ACCIDENTS: THREE LESSONS

Accident	System	Feedback	Outcome
Slotin (1946)	Bare Pu core	None	Prompt critical $\rightarrow$ instant lethal dose. Human reaction time irrelevant.
BORAX (1954)	BWR with UO_2 fuel	Negative (fuel temp.)	Feedback stopped reaction, but fuel melted first $\rightarrow$ steam explosion.
Chernobyl (1986)	RBMK reactor	**Positive** (steam-void)	Feedback *amplified* excursion $\rightarrow$ catastrophic destruction.

The Universal Rule:

- Prompt criticality must be prevented by design
- Negative feedback helps, but isn't fast enough to prevent damage
- Positive feedback is absolutely unacceptable
- Multiple independent safety systems are essential

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AI-Know Platform:

- Secure, compliant AI environment aligned with NUS policies
- Conversational AI for educational content development
- Private interactions that are never used to train models

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For more information about AI-Know: https://ai-know.nus.edu.sg

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